SILO

Conceptual Architecture

(Pre-draft)

WHAT IS SILO?

SILO defines a common semantic model and interoperability layer that enables consistent understanding and exchange of infrastructure operational data across vendors, systems and management domains.

- **Interoperability**
  Works across any vendor, any protocol.
- **Efficiency**
  Enables smarter operations and lower energy use.
- **Innovation**
  Accelerates new applications and services.

Green Software Foundation

* This is a draft architecture for discussion and evolution.

Pre-draft proposal – Nov 2026

APPLICATIONS & CONSUMERS

DCIM / Infrastructure Management | Workload Orchestrators | Digital Twins & Simulation | Energy Management & Optimization | Analytics & AI/ML Applications

SILO SEMANTIC LAYER

Common Information Model

A common semantic model that standardizes the meaning of operational data across the data center infrastructure lifecycle.

MAPPING & TRANSLATION

Maps source schemas and protocols to the SILO semantic model.

SILO APIS

Standard APIs for secure access to normalized data and services.

PROTOCOL ADAPTERS

Redfish | BACnet | Modbus | SNMP | MQTT | OpenTelemetry | ...

INFRASTRUCTURE & DATA SOURCES

Compute | Cooling | Power | Facility | Network | Storage | Telemetry & Events

DESIGN PRINCIPLES

- Vendor neutral
- Complements, not replaces WDPC
- Open, extensible & future-proof
- Secure by design
- Community-driven

STANDARDS ECOSYSTEM

- **WDPC (GSF)** — Workload Dynamic Power & Cooling
- **Redfish (DMTF)** — Device management
- **OpenTelemetry** — Observability
- **SNIA Swordfish** — Storage management (DMTF Redfish extension)
- **OCP Hardware Management** — Open Compute Project
- **OCP Cooling Environments** — Open Compute Project
- **Others** — IEEE, ASHRAE, IEC, ...

SILO maps to and leverages existing standards.